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Calculation of low-energy elastic cross sections for electron-CF₄ scattering

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A new computational approach has been used to evaluate the rotationally summed, vibronically elastic integral cross sections from the scattering of slow electrons (energy ranging from 1.0 eV up to 40.0 eV) by tetrafluoromethane molecules in the gas phase. The various symmetry components have been analyzed using the exact static exchange approximation and also by including a nonempirical, model polarization potential employed before by our group. A comparison with earlier calculations and with existing experiments allows us to assign the symmetries of the shape resonances in the 5–30 eV energy region which are seen by experiments and are also shown by the present calculations.

I. INTRODUCTION

The series of fluorinated methanes constitute one of the chief molecular systems, among the many possible candidates, which are of interest in various technical applications such as plasma etching, the setting up of discharge switches and of gas-filled particle detectors.^{1,2} In all of them, the study of the interaction of low-energy electrons with the respective gas molecules is a critical element for the definition of the properties of the devices under analysis. Consequently, knowledge of the cross sections for the basic processes (e.g., elastic and inelastic scattering, dissociative attachment, impact dissociation and fragmentation, and electronic direct ionisation, just to mention a few of them) allows the tailoring of gas mixing and of operating conditions to a particular purpose and to a selective enhancement of some of the possible processes.

The first molecule of the series, the fully fluorinated methane derivative, CF₄, has therefore been the object of a large variety of studies over the years, both from the theoretical and experimental sides. For example, integral cross sections for elastic and inelastic scattering have been derived from swarm data by several authors^{3,4} and, although considerable discrepancies were found at a more quantitative level, all results indicated the existence of a minimum in the elastic cross sections at energies below $E=0.5$ eV and the presence of strong vibrational excitations at the low energies. Field *et al.*⁵ gathered similar conclusions for their electron transmission experiments with energies between 50 and 450 meV.

The total cross sections for all molecules in the series of fluoro-chloromethanes were measured in electron transmission experiments by Jones⁶ for impact energies between 1 and 50 eV. Those pertaining to our title molecule, CF₄, displayed a very broad shoulder around 3.6 eV, a marked peak at 8.9 eV, followed by a very broad, lower maximum around 26 eV, with a monotonic decrease of the cross sections as the collision energy increased up to 50 eV.

Further measurements of differential cross sections (DCS) were recently provided for vibrationally elastic and inelastic scattering processes⁷ in the energy range of 0.3–20 eV. This last set of experiments was also extended to study the resonant vibrational excitation in electron-CF₄ collisions⁸ and the authors found that, in such a system, the inelastic scattering becomes comparable in magnitude with elastic scattering chiefly in two regions: below 2 eV and between 6 and 11 eV.

From the computational side, an early estimate of the energetic positions of the possible low-energy shape resonances came from the continuum-multiple-scattering (CMS)- $X\alpha$ model calculations of Tossell and Davenport,⁹ whereby the authors provided a “semiquantitative” model cross section for elastic electron-CF₄ scattering while also observing that the existing structures markedly depended on the chosen geometry of the target. They assigned to the t_2 symmetry the narrow resonance seen⁶ around 3.6 eV and also predicted a Ramsauer–Townsend (RT) type of minimum around 1 eV.

Further calculations were carried out only recently,¹⁰ at somewhat higher energies ($5 \text{ eV} \leq E \leq 35 \text{ eV}$) by applying the multichannel Schwinger variational principle (SMC) in the static exchange (SE) approximation and for elastic scattering processes. The resulting DCS could not be compared to experiments since none were available at the chosen collision energies, while the calculated partial integral cross sections for each irreducible representation (IR) of the target molecule were used to determine positions and symmetries of the several resonances which showed up in the calculations as maxima in the elastic cross sections. The latter results suggested the existence of four broad shape resonances in the fixed-nuclei-approximation (FNA) approach: two of t_2 symmetry, one of a_1 symmetry and one of e symmetry.

More recently, the same computational approach has been independently used¹¹ to obtain elastic electron-CF₄ scattering cross sections from 1 to 40 eV. These authors employed a more extensive variational basis set than that used in Ref. 10 and incorporated in the code certain computation

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improvements that should lead to an increased accuracy of the final results. The approach still did not include polarization effects but produced, above 10 eV, a better general agreement with the measurements of Ref. 6, although still finding marked discrepancies with the elastic results from the measurements of Mann and Linder⁷ and of Boesten *et al.*¹² The latter measurements, on the other hand, are also quite different from the former at collision energies above 5 eV. It seems that the difference chiefly arises from the needed extrapolation of the measured DCS into the forward scattering and backscattering regions which are experimentally inaccessible and therefore a final experimental answer about this point is not yet available. It is interesting to note, however, that the measurements of Ref. 12 are higher than the ones given by Ref. 7 at energies above 5 eV and get closer to the experiments of Jones⁶ around 15 eV.

The questions posed by the experiments described earlier, therefore, have been only partially answered by the existing calculations, which have not yet been carried out with the inclusion of a description of the response function for the target system. In particular, experiments and calculations suggest the following.

(i) Integral elastic cross sections decrease as the collision energy goes down to 1 eV and below.

(ii) The maxima in the elastic and inelastic cross sections as the energy goes up to about 40 eV, from the lowest available data⁵ indicate the presence of more than one negative-ion temporary state for the CF₄ molecule, all being possibly related to open-channel resonant scattering.

(iii) The existing SE calculations still disagree with each other, although they indicate the presence of at least one low-energy resonance of t_2 symmetry, of a further resonance of a_1 symmetry located in the vicinity of the previous one and of a broader, less intense resonance of e symmetry located above 20 eV.

It therefore becomes of interest to try and compute once more the elastic cross sections, within the same range of energies, but using an entirely different computational algorithm which could also test the effect of polarization forces on the computed cross sections. This will be the task of the present work, where we have carried out exact SE calculations of elastic integrated cross sections and further included model polarisation forces to describe the molecular response to the field induced by the impinging electron (SEP calculations). Section II reports the general features of our theoretical model and of our computational algorithm, while Sec. III shows our results and compares them with the experiments and the other calculations discussed before. Our final conclusions are summarized in Sec. IV.

II. THE COMPUTATIONAL MODEL

Since the details of the general formulation for the scattering of an electron by an N -electron nonlinear molecule, containing M nuclei, have been derived and presented in some of our earlier publications,^{13–17} we will report in this section only a brief outline of the method and discuss our more recent computational implementation of it.

We describe the collisional process in terms of the solution of the Schrödinger (in a.u.) of the form

$$H\psi(r, X) = E\psi(r, X), \quad (1)$$

where

$$H = \hat{T} + \hat{V} + \hat{H}_{\text{mol}} \quad (2)$$

with $\hat{T}$ the kinetic energy of the incident electron, $\hat{V}$ its interaction with the molecular nuclei and electrons,

$$\hat{V} = \sum_{i=1}^N (r - x_i)^{-1} + \sum_{\gamma=1}^M Z_{\gamma}(r - X_{\gamma})^{-1}, \quad (3)$$

$\hat{H}_{\text{mol}}$ the Hamiltonian for the molecular target, and r the position of the continuum electron. We let X represent, collectively, the coordinates of the target electrons x_i ($i = 1, N$) and of the molecular nuclei X_{γ} ($\gamma = 1, M$) and intend to refer all particles to a frame of reference fixed to the molecule (BF frame). The molecular Hamiltonian, $\hat{H}_{\text{mol}}$, describes not only the interaction among the electrons and nuclei, but also their respective motion. One now converts the many-body problem to an effective single-particle one by expanding the total system wave function ψ in terms of a complete set of target states through which one obtains the familiar set of coupled integrodifferential equations¹³ (IDE's),

$$H_{\alpha} F_{\alpha}(r) = \sum_{\beta} Z_{\alpha\beta}(r), \quad (4)$$

which constitute the general close-coupling (CC) equations of the scattering problem, with the following meaning of symbols:

$$H_{\alpha} = (\nabla^2 + k_{\alpha}^2), \quad (5a)$$

$$Z_{\alpha\beta}(r) = \int K_{\alpha\beta}(r|r_1) F_{\beta}(r_1) dr_1, \quad (5b)$$

$$K_{\alpha\beta}(r|r_1) = V_{\alpha\beta}(r_1) \delta(r - r_1) + W_{\alpha\beta}(r|r_1). \quad (5c)$$

Here k_{α}^2 is given by $2(E - \epsilon_{\alpha})$. The direct electrostatic interaction, $V_{\alpha\beta}$, is local while the exchange interaction is schematically represented by $W_{\alpha\beta}$, a complicated, nonlocal term.

The simplest *ab initio* treatment of Eq. (4) is to evaluate the direct and nonlocal interactions of Eqs. (5a) and (5b) without any further polarization effects. This is the so-called exact static exchange level (ESE).¹⁴ The further addition of short-range correlation effects between the bound and the continuum electrons, together with the long-range part of the polarization forces, requires the selection of a local energy-independent correlation-polarization (CP) model potential, $V_{\text{CP}}(r)$. We have previously discussed such a potential by defining an average correlation energy of a single particle, within the formalism of the Kohn and Sham variational theorem, and by obtaining the short-range correlation forces as an analytic function of the target electronic density $\rho_{\text{HF}}(r_i)$.¹⁷ Such an analytic function is then matched in the long-range region to the second-order perturbation expansion for the polarization potential

$$V_{\text{pol}}^{(2)}(r) \sim - \sum_{l=1}^{\infty} \frac{a_l}{2r^{2l+2}}, \quad (6)$$

where usually only the first multipolar coefficient, the target static polarizability α_D , is employed. This will be called the exact static exchange plus polarization (ESEP) approach.

In the numerical method, employed to solve Eq. (4) we then convert the CC equations into a set of coupled radial equations by making first a single center expansion (SCE) of the bound and continuum functions and then integrating over all the angular coordinates. Such an approach is usually rather inefficient for the evaluation of the electronic properties of molecular systems since the form of the relevant wave function near the nuclei is critical for determining the total energy and other related observables. For scattering problems, however, the nuclear singularities are relatively less important¹⁴ and, with proper care, the SCE approach can yield highly reliable results as we shall see later.

One important point in our approach is the construction of the symmetry adapted, generalized harmonics X as linear combinations of spherical harmonics $Y_l^m(\theta, \phi)$ which, for given l , form a basis of the $(2l+1)$ dimensional IR of the full rotation group. If such generalized harmonics satisfy the usual orthonormality relations, then the coefficients over the Y 's correspond to a unitary transformation between the X' and the spherical harmonics¹³

$$X_{hl}^{\mu}(\theta, \phi) = \sum_m b_{hlm}^{\mu} Y_l^m(\theta, \phi). \quad (7)$$

In the SCE approach expansion all functions are therefore written as

$$f(r, \theta, \phi) = \sum_{l,m} f_{lm}(r) Y_{lm}(\theta, \phi). \quad (8)$$

The product of such two functions could be given by first transforming the angular momentum representation of f into a coordinate representation using

$$f_{\alpha\beta}(r) = \sum_{l,m} f_{lm}(r) U_{lm,\alpha\beta}, \quad (9)$$

where $U_{lm,\alpha\beta} = Y_{lm}(\theta_\alpha, \phi_\beta)$ and $f_{\alpha\beta}(r) = f(r, \theta_\alpha, \phi_\beta)$. In this representation a product of two functions is just a point-by-point product of the form¹⁸

$$(fg)_{\alpha\beta}(r) = f_{\alpha\beta}(r) g_{\alpha\beta}(r). \quad (10)$$

The angular momentum representation can then be recovered by transforming back from the coordinate representation

$$f_{lm}(r) = \sum_{\alpha,\beta} f_{\alpha\beta}(r) V_{\alpha\beta,lm}, \quad (11)$$

where

$$V_{\alpha\beta,lm} = Y_{lm}(\theta_\alpha, \phi_\beta) W_\alpha W'_\beta. \quad (12)$$

The weights and points for the θ integration are those given by the Gauss-Legendre integration formulas so that for an n point integration, the point θ_α is the α th zero of the Legendre polynomial $P_n(\cos \theta)$ and the weight W_α is given by

$$W_\alpha = \frac{-2}{(n+1)P'_n(\cos \theta_\alpha)P_{n+1}(\cos \theta_\alpha)}. \quad (13)$$

The weights and points for the ϕ integration are those given by the Gauss-Chebyshev integration formulas applied once to the $0 \rightarrow \pi$ region and once to the $\pi \rightarrow 2\pi$ region so that for a $2n$ point integration in the region $0 \rightarrow 2\pi$, the weight W'_β is given by

$$W'_\beta = \frac{\pi}{n}, \quad (14)$$

and the point ϕ_β is given by

$$\phi_\beta = \frac{(2\beta-1)\pi}{2n}. \quad (15)$$

The conventional evaluation of Eq. (10) requires a computational effort proportional to the third power of the number of partial waves. However, the computational effort in the transformation method given by Eq. (9) depends only on the second power of the number of partial waves. Thus, for larger molecules this approach can significantly reduce the effort needed to evaluate the required exchange matrix elements.

In our polyatomic code this quadrature is used to perform all angular integrals. However, when the quadrature is not exact, as in the case of the partial wave expansion of the orbitals, of the local exchange potential, and of the V_{CP} potential, then the number of angular points can be increased until satisfactory convergence is obtained.

Because of the importance of correctly describing the short-range part of the exchange interaction of Eq. (5c), we have implemented in the present treatment an iterative-exchange procedure described and applied¹⁹ earlier to other polyatomic systems. Briefly, the method requires the evaluation of the following sum of integrals:

$$\sum_\alpha \int \phi_\alpha(r') |r-r'|^{-1} F^{(p\mu)}(r') dr' \phi_\alpha(r), \quad (16)$$

where α sums over the occupied target molecular orbitals (MO's), given by the ϕ_α functions, and the $F^{(p\mu)}$ are the continuum electron functions for any IR labeled by the $|\pi p \mu\rangle$ indices mentioned earlier. The corresponding matrix formulation of the CC equations (4) can be written as

$$L F_i^{(p\mu)} = W F_{i-1}^{(p\mu)} \quad (17)$$

where $W F^{(p\mu)}$ is the exchange term given by some initial estimate for $i=1$, $F_0^{(p\mu)}$, and the iterative procedure consists in generating the successive $F_i^{(p\mu)}$ matrices until convergence is achieved for the unchanged structure of the scattering K matrix within a given threshold of invariance for its matrix elements. Such a procedure usually works rather slowly and, even with the accelerating devices suggested earlier by us,¹⁹ still requires several iterations (between 20 and 30) before satisfactory convergence is reached. We have therefore further improved our procedure by taking advantage of the Schwinger variational method applied to molecular collisions by McKoy *et al.*,²⁰ as briefly described later.

The equations are rewritten in terms of a Lippmann–Schwinger equation for the scattering by a potential which is the difference between the local exchange potential and the exact exchange potential

$$\psi_p = \phi_p + G_\alpha V_e \psi_p, \quad (18)$$

where ϕ_p is one of the solutions of a purely local potential

$$H_\alpha \phi_p = E \phi_p \quad (19)$$

in which the V_α potential is the sum of local contributions

$$V_\alpha = V_{\text{static}} + V_{\text{polarization}} + V_{\text{model exchange}}. \quad (20)$$

The potential V_e is defined by

$$V_e = V_{\text{exact exchange}} - V_{\text{model exchange}} \quad (21)$$

and the Green's function is defined by

$$(E - H_\alpha) G_\alpha = 1. \quad (22)$$

If we define the total K matrix by

$$K_{pq} = K_{pq}^{(\alpha)} + K_{pq}^{(e)}, \quad (23)$$

where $K^{(\alpha)}$ is the K matrix due scattering only by V_α , then $K^{(e)}$ can be obtained as

$$K_{pq}^{(e)} = -2 \langle \phi_p | V_e | \psi_q \rangle. \quad (24)$$

We then evaluated the K matrix by making use of Pade's approximants in a way very similar to that developed by Lucchese and McKoy for linear molecules.²¹ To obtain the corresponding Pade's approximants, the approach starts from some arbitrary function ψ to evaluate the further function $G_\alpha V_e \psi$. Such a treatment only required a rather small modification of the asymptotic boundary conditions in our MATCH subroutine. One can then define a sequence of functions $\phi_p^{(i)}$ which are given by

$$\phi_p^{(i)} = \langle G_\alpha V_e \rangle^i \phi_p \quad (25)$$

such that $\phi_p^{(0)} = \phi_p$ is just the solution of the purely local problem. Then the $[N/N]$ Pade's approximant can be obtained from the expression²²

$$K_{pq}^{(e)}[N/N] = \sum_{i,j=0,N-1} \langle \phi_p | V_e | \phi_q^{(i)} \rangle D_{ij}^{-1} \langle \phi_p | V_e | \phi_q \rangle, \quad (26)$$

where D_{ij}^{-1} is the (i,j) th element of D^{-1} , which is the inverse of the matrix D with elements

$$D_{ij} = \langle \phi_p^{(i)} | V_e - V_e G_\alpha V_e | \phi_q^{(j)} \rangle. \quad (27)$$

Such an approach was found to converge very quickly and to provide the same answers given by the previous methods, with a marked saving of computational time.²³

III. CALCULATION OF ELASTIC CROSS SECTIONS

In order to carry out the present calculations for the title molecule, we started with self-consistent field (SCF) target multicenter wave function expanded over a double-zeta-plus-polarization (DZP) set of 80 GTO basis functions, using a conventional contraction scheme of $9s5p1d/4s2p1d$ both on the F and C atoms. The tetrahedral (T_d) symmetry of the target was employed within the fixed-nuclei approximation

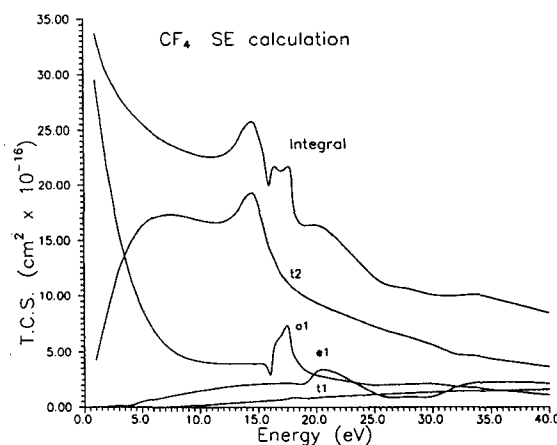


FIG. 1. Computed elastic integral cross sections, partial and total, for electron scattering from CF_4 . The TCS (total cross sections) are given in 10^{-16} cm^2 units and correspond to exact static exchange (SE) calculations. The symmetry notation is given along each curve.

(FNA) scheme, with a bond distance of $2.49206a_0$. The calculations yielded a total energy of -435.76699 a.u. , better than that of Ref. 10 and close to the result of Ref. 11. For the long-range part of the polarization potential we employed the spherical component of the experimental dipole polarizability, α_0 , which was $25.9a_0^3$, as previously used by us,²⁴ for the same system in carrying out effective potential calculations.

The SCE procedure was numerically implemented by expanding the occupied MO's over a (r, ϑ, φ) grid of $(208 \times 56 \times 26)$, for a total 302 848 points. The actual scattering calculations for solving the coupled radial equations were carried out on an r grid of 416 points, out to a matching distance of $\sim 12a_0$ which ensured that the static potential had essentially vanished and that the polarization contribution was of the order of $\sim 10^{-3} \text{ a.u.}$

The multipolar expansion of the total target density was carried out up to $l_{\text{max}1} = 15\hbar$, while the coupling potential in the scattering equations was expanded up to $l_{\text{max}2} = 30\hbar$, i.e., much larger values than those used by us for the CH_4 molecule,²⁴ where convergence tests indicated that $l_{\text{max}2} = 12$ was already sufficient to reach converged K -matrix elements.

We report in Fig. 1 our exact SE calculations, showing both the total integrated cross section and the partial contributions from the significant IR involved in the scattering process. Because of the size of the expansion and of the number of points used in the SCE calculations, we did not probe the very low-energy room-temperature Ramsauer–Townsend (RT) region below 1000 meV but concentrated instead on a study of the shape resonances above 3 eV and up to 40 eV.

It is interesting to note how close the present SE results are to the SMC multicenter SE calculations of Winstead *et al.*¹¹ who show (in Fig. 2 of their work) very similar cross sections, in shape and in order of magnitude, to those we report in Fig. 1. The results of our SCE calculations clearly indicate the following features of the integrated, rotationally summed, FNA cross sections.

(i) A narrow shape resonance of t_2 symmetry below 15 eV of collision energy.

(ii) A broad "hump" in the t_2 contribution at lower energies, with its maximum around 5 eV.

(iii) A rapidly decreasing a_1 contribution, as the energy increases, with the presence of another narrow resonance at energies above 15 eV. This feature is responsible for the double maximum appearing in our total elastic cross sections.

(iv) a broad maximum in the e component of the cross section above 20 eV, indicating the presence of a further shape resonance of that symmetry in the higher-energy region of the integrated elastic scattering.

Our results, although differing from those reported earlier in Ref. 10, are close to the more recent SMC calculations of Ref. 11 which indeed exhibit the same general features and therefore indicate strong similarities among theoretical results at the SE level. It is also reassuring to note that, in a more recent set of calculations (Ref. 25), the group of Dr. Huo has obtained SE cross sections which are once more very close to the present results and to those of Ref. 11. Thus, we can say now that three different and independent calculations at the SE level produce cross sections which are in good accord with each other and which exhibit, in the region between 5 and 30 eV, the three types of shape resonances mentioned earlier.

Such calculations are, however, still in disagreement with the experimental findings, as they all surmise elastic cross sections which, in the strongest resonance region ($5 \leq E \leq 15$ eV), are larger than those from the experiments of Mann and Linder⁷ and of Boesten *et al.*¹² and are more in accord with the measured total cross sections of Jones.⁶ Moreover, all SE calculations report resonance positions at higher energies than those indicated by the experiments.

We have therefore carried out further calculations which included a description of the response function of the electronic target density to the perturbing field of the incoming electron. In these calculations, as briefly discussed in Sec. II, we have made use of a nonempirical model potential which treats the short-range part of the correlation effect via a density functional (DF) form derived earlier on for a free-electron gas description of the target electrons.^{26,27} The long-range region of the same interaction is then treated via second-order perturbation theory²⁸ and employs the spherical component of the dipole polarizability of the title molecule.

What one qualitatively expects to happen when short-range and long-range polarization forces are included is that the open-channel resonances should move to lower collision energies if they are already strong enough at the SE level, or that they could become indistinguishable from the contribution of the background cross section for that particular symmetry if they are weaker features of the static exchange calculations over a fast varying background contribution.

The results of our SE results including polarization effects (SEP computations) are shown in Fig. 2. A comparison between the computed results of our two Figures indicates the following.

(i) The resonance of t_2 symmetry below 15 eV is still rather marked, having now moved to lower collision ener-

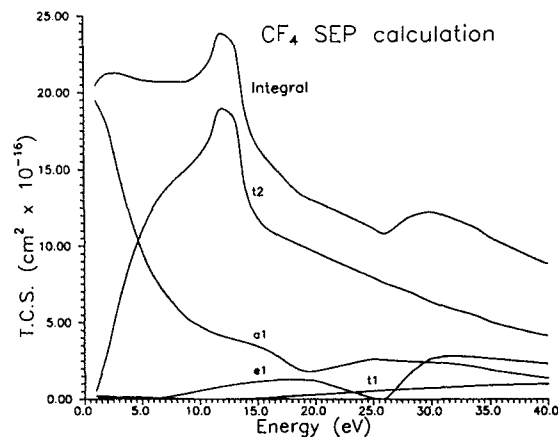


FIG. 2. Same as in Fig. 1 but with the inclusion of a model description of the target response function (SEP calculations). The notation is the same as in Fig. 1.

gies. Thus, this shape resonance is now around 10 eV and very close to experiments, as we shall discuss later.

(ii) The nearby resonance of a_1 symmetry has now become less visible as the contribution from the a_1 partial cross section shows a rapidly varying background which increases at low energies, albeit less strongly than before. As a result, the total elastic cross section now exhibits a broad "hump" in the energy range around 1 eV, somewhat closer to the experiments.⁶

(iii) The third resonance at higher collision energy, shown in the SE calculations to be of e symmetry, is still present in the SEP calculations but has now moved to higher energies, i.e., around 30 eV, where it has become more marked than that shown by the earlier SE calculations.

For reasons of clarity we have further reported in Fig. 3 a comparison between total elastic integral cross sections computed in the present work at the SE level (dashed line) and at the SEP level (solid line). It is very interesting to see that our model treatment of the target response function ap-

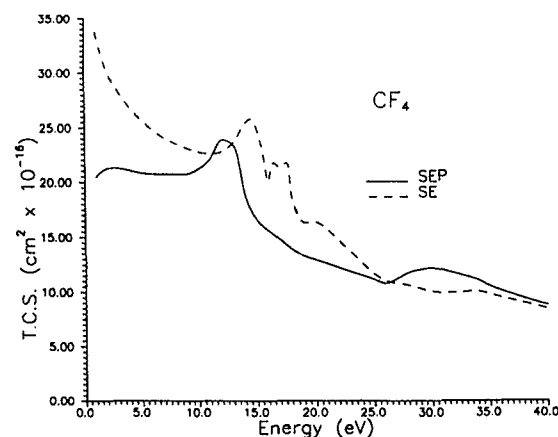


FIG. 3. A comparison of integral, total elastic cross sections computed in the present work. The dashed line refers to the exact static exchange calculations, while the solid line describes calculations which also include polarization effects.

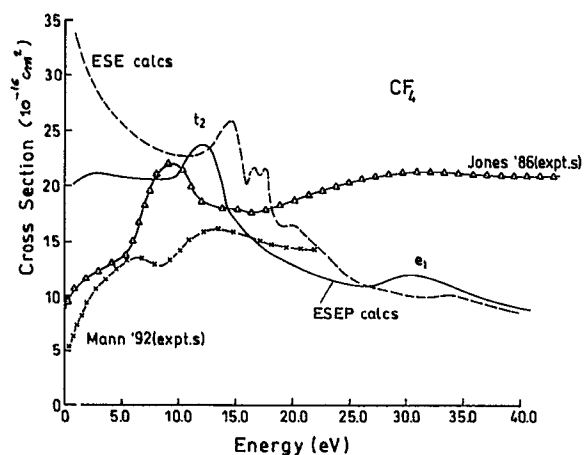


FIG. 4. A comparison of computed and measured elastic integral cross sections. Dashed line: present exact static exchange (ESE) calculations. Solid line: present calculations including polarization effects (ESEP). Open triangles: experimental total integral cross sections from Ref. 6. Crosses: experimental elastic integrated cross sections from Ref. 7.

pears to provide computed results which improve agreement with experiments and which change according to physical intuition. This has indeed been the case in all our previous calculations of polyatomic targets^{23,24,29-31} and give us some level of confidence on the use of such a model.

The SEP results, in fact, now indicate the presence of a single maximum around 10 eV, as indeed shown by Jones' experiments.⁶ Moreover, they show now a much flatter behavior as E decreases and present a broad "hump" around 1 eV, as also indicated by the earlier experiments. They also suggest a further resonance of e symmetry in the energy region around 30 eV. Finally, the general behavior of the SE cross section above 15 eV, as a decreasing function of collision energy is also confirmed by the SEP calculations.

A more explicit comparison between the SE and the SEP calculations of the present work with experiments is reported in Fig. 4, where no normalization between theory and experiments has been performed as absolute values were available for the measured data.^{6,7} One clearly sees in Fig. 4 how well our SEP calculations are faring in comparison with the existing experiments: the single resonance indicated by the total measurements of Jones⁶ is well described and its broad maximum around 30 eV is also given by our elastic cross sections. The discrepancy in size above the 15 eV range of energy can be attributed to the fact that the measured values⁶ also include vibrational and electronic excitations. In an attempt to estimate the inelastic contributions, Winstead *et al.*¹¹ showed that the SE calculations should come out to contribute nearly 50% of the measured values, which is in keeping with our results. Moreover, one generally expects polarization effects to be less important as the energy increases. The fact that the SEP cross sections are seen here to differ little from SE results, as far as the background scattering is concerned, is once more in accord with physical intuition.

The agreement between our computed location and size of the t_2 resonance with the experiments can be reconciled

with the total cross section measurements⁶ if one assumes that almost all of the resonant scattering leads to vibrational excitation. The evidence from other experiments strongly suggests that the most effective inelastic process is the excitation of the ν_3 mode (asymmetric stretch) whose cross section has a peak around 9–10 eV and whose angular distribution is consistent with a shape resonance of t_2 symmetry.^{8,12}

The difficulty in locating an a_1 resonance when polarization forces are included may also be due to the fact that our calculations show, for this symmetry, a very rapid variation of the background cross section as the energy decreases and, therefore, a search for the resonance would require an analysis of the eigenphase sum over a rather narrow range of energies. This will be the object of further study, together with an analysis of angular distributions at the available energies.⁷

Finally, the SEP calculations eliminate the spurious increase of SE total cross sections with decreasing energy and show instead an overall slow decrease, in qualitative accord with the experiments. Here again, the low-energy behavior of the A_1 component needs to be more carefully analyzed in order to clarify the physical origins of the discrepancy.

A further comparison of our calculations with previous calculations and with experiments is shown in Fig. 5, which therefore summarizes our present findings. We report in Fig. 5 the total cross sections data⁶ (open circles), the elastic measurements of Mann and Linder⁷ (open triangles), and those of Boesten *et al.*¹² (open squares). The previous calculations are all at the SE level and are given by the dotted curve,¹⁰ the dashed curve,¹¹ and the solid line.²⁵ Our calculations are at the SEP level and are given by the dotted-dashed curve.

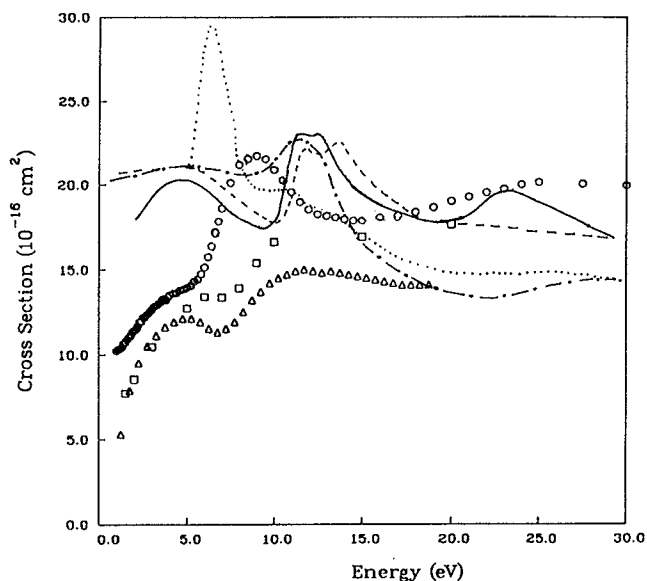


FIG. 5. A comparison of present results with previous calculations and experiments. Experimental data: Open circles from Ref. 6; open triangles from Ref. 7; open squares from Ref. 12. ESE calculations: Exact static exchange elastic cross sections from Ref. 10 (dotted curve), from Ref. 11 (dashed curve), from Ref. 25 (solid line). The present ESEP calculations are given by the dotted-dashed curve.

TABLE I. Total elastic integrated cross sections computed in the present work using exact static exchange calculations (ESE) or exact static exchange plus model polarization (ESEP) calculations. All values are in units of 10^{-16} cm².

Energy (eV)	ESE cross sections	ESEP cross sections
1.0	33.79	20.46
2.0	30.23	21.23
3.0	28.24	21.23
4.0	26.77	21.02
5.0	25.66	20.80
6.0	24.77	20.71
7.0	24.04	20.69
8.0	23.47	20.69
9.0	23.02	20.71
10.0	22.68	21.15
11.0	22.55	22.10
12.0	22.73	23.87
13.0	23.60	23.37
14.0	25.38	18.91
14.5	25.78	17.51
15.0	24.84	16.64
15.5	22.82	16.06
16.0	20.00	15.62
16.5	21.73	15.22
17.0	21.30	14.80
17.5	21.67	
18.0	18.75	13.96
19.0	16.34	13.29
20.0	16.38	12.95
25.0	11.71	11.18
26.0	11.06	10.81
28.0	10.63	11.82
30.0	10.09	12.23
32.0	9.99	11.75
34.0	10.10	11.10
35.0	9.85	10.57
40.0	8.49	8.86

One clearly sees that the present results describe the strong resonance rather well and behave similarly to the other calculations in the low-energy region. Moreover, they also indicate the presence of a broad resonance above 25 eV, as surmised by experiments.⁶ Table I gives the actual values of the computed total elastic cross sections discussed in this section.

IV. CONCLUDING REMARKS

In this work we have presented a new set of *ab initio* calculations for low-energy electron scattering from CF₄ molecules. The theoretical treatment used an SCE approach and treated exactly the static and the exchange interactions during the scattering process. Furthermore, we carried out additional calculations which included a model description of short-range polarization forces while correctly treating the leading term of the dipole polarizability long-range coefficient.

Our SE results generally agree with previous calculations and surmise the presence of open channel resonances of t_2 , a_1 , and e symmetries in the energy range between 5 and 40 eV. The SEP calculations confirm the presence of a strong shape resonance of t_2 symmetry around 10 eV, thus being in

agreement with total cross section measurements.⁶ They also indicate the following features of the computed cross sections.

(i) A slow decrease of the elastic cross sections in the low-energy region below the t_2 resonance in general accord with existing experiments.

(ii) A large contribution of the inelastic channels for energies above 15 eV, in accord with purely elastic data⁷ in that energy region.

(iii) A broad resonance of e symmetry in the energy range above 26 eV.

(iv) A strong increase, with decreasing energy, of the a_1 contribution to the scattering process. This feature exists also in our SE calculations and in the previous SMC calculations at the SE level,¹¹ where calculations suggest a further shape resonance close in position to the t_2 resonance.

It is also interesting to note that recent SEP calculations by Huo's group²⁵ also suggest the presence, around 10 eV, of two superimposed resonances of t_2 and a_1 symmetries and a broad increase of the cross sections above 20 eV, as shown by our SEP results.

The present discussion further indicates that the analysis of CF₄ elastic cross sections, integral and differential, still requires more calculations and more extensive comparisons with measurements, especially in the low-energy region where correlation effects play a crucial role. The actual presence of an RT minimum below 1 eV still needs to be verified by calculations and the discrepancies existing between various measurements also need to be explained.

The current level of understanding is, however, already significant for some of the features found by the experiments, like the possible symmetries of the open-channel resonances and the general behavior of integrated cross sections above 5 eV. The present calculations are now the third independent set of computations on the title system and confirm the origin of several of the observed features. They also give, for the first time, indications on the possible effects coming from the inclusion of target response functions in the scattering calculations.

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